

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 2.2: Performance of Classification Algorithms

Jana Lasser

Research group

Complex Social & Computational Systems (CS)²

IDEa_Lab

Das Interdisziplinäre Digitale
Labor der Universität Graz



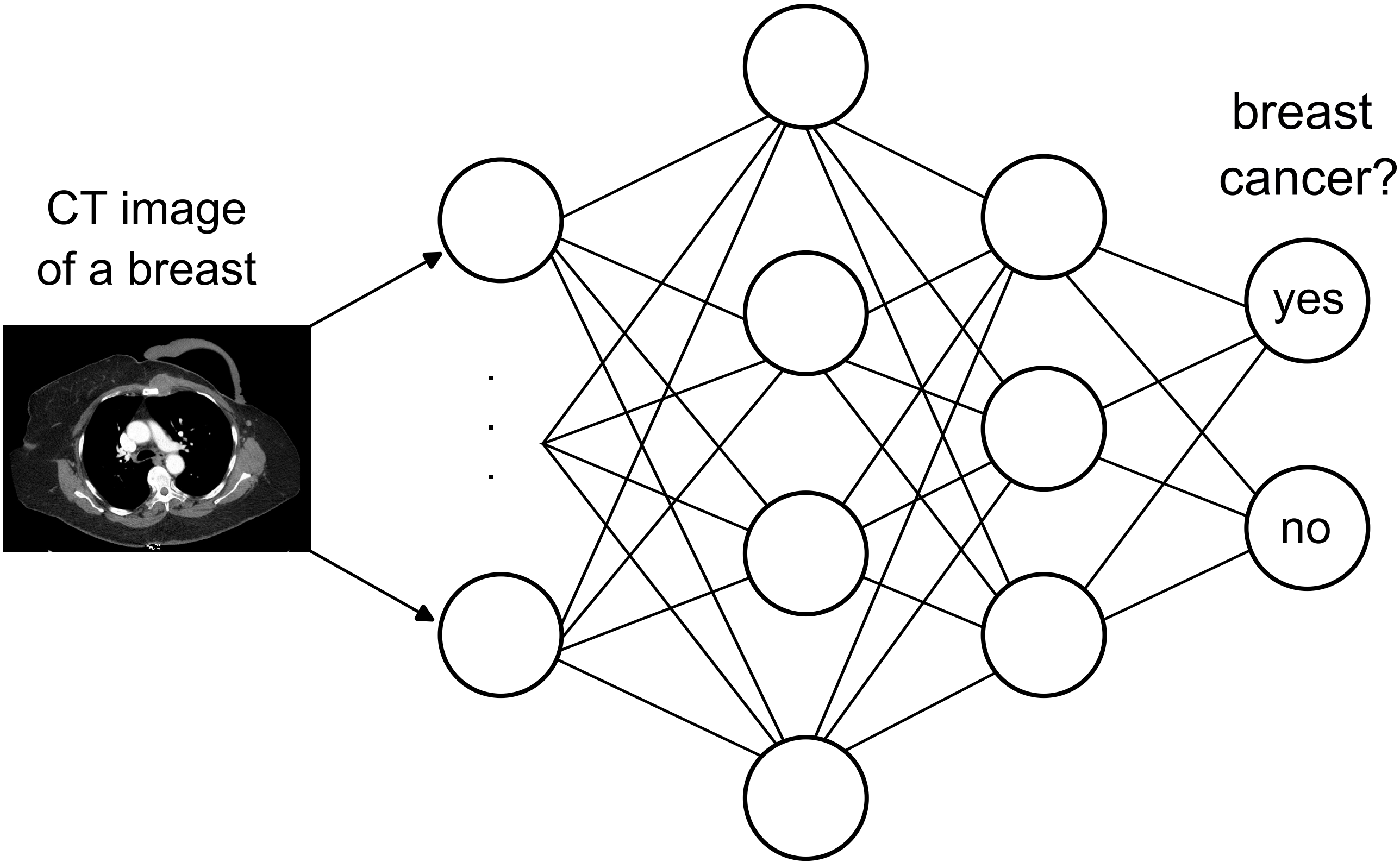
Performance

- The **performance** of a classification algorithm measures how well it can assign observations x to classes y .
- Knowing an algorithm's performance is important at three points:
 1. To **guide training** towards better performance.
 2. To **stop training** when performance no longer improves.
 3. After training ends, to assess the algorithm's **final performance**.

*Next, we'll learn different ways to **measure** an algorithm's performance.*

Example: Breast Cancer Classification

"yes" = cancer present
"no" = cancer not present



TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Source example image breast cancer: [Wikimedia Commons](#), CC BY-SA 3.0.

Classifier Performance

"yes" = cancer present
"no" = cancer not present

Naive approach: Calculate the error rate.

Error rate: Proportion of observations that were incorrectly classified.

Here: 14 observations, 4 errors = 28.6% error rate.
Or $100\% - 28.6\% = 71.4\%$ **accuracy**.

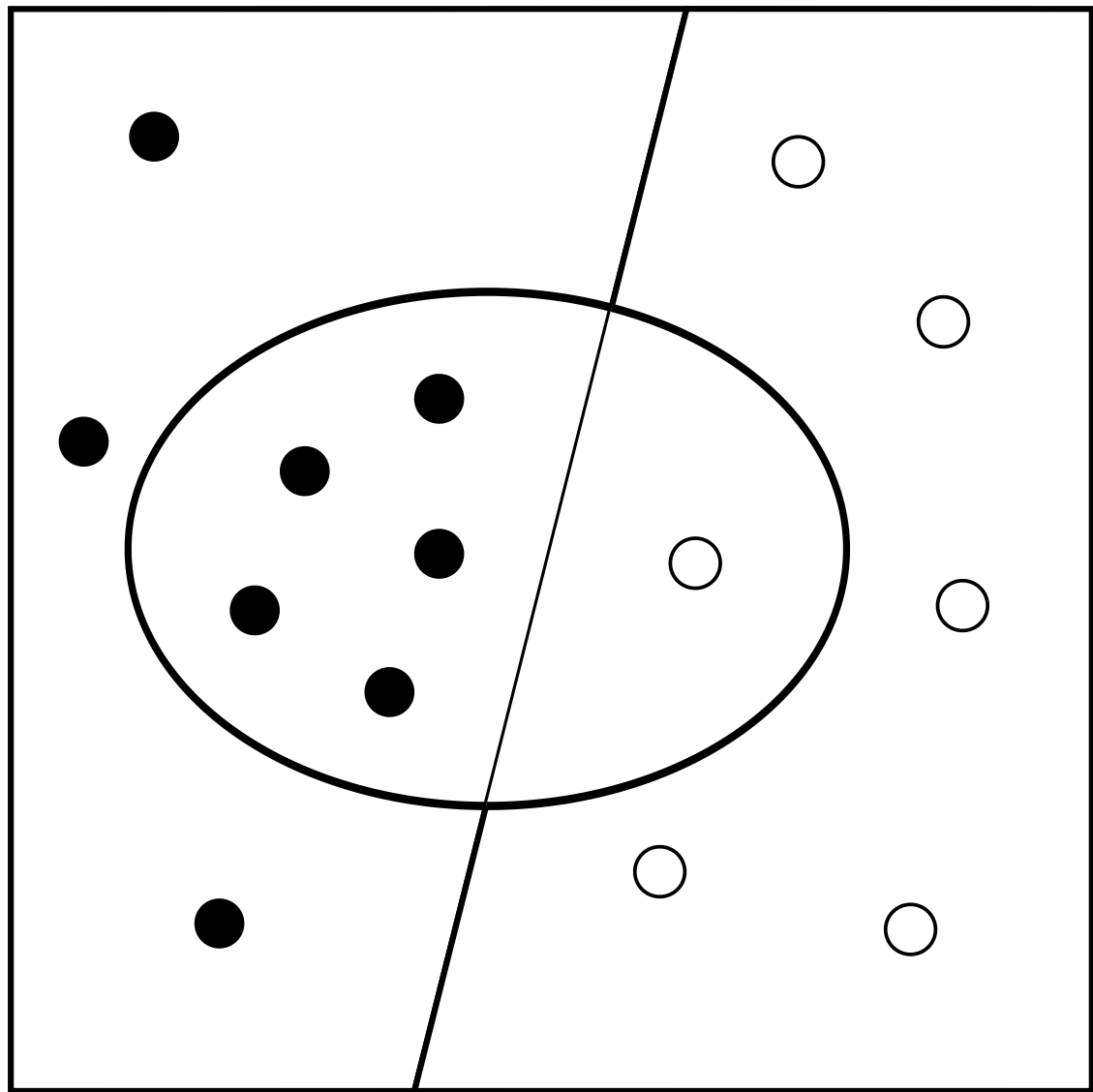
Problem: The classifier can make **two types** of errors that are indistinguishable here:

- 1. **False positive:** cancer is detected in an image even though none is present.
- 2. **False negative:** cancer is not detected in an image that contains it.

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

"yes" = cancer present
"no" = cancer not present

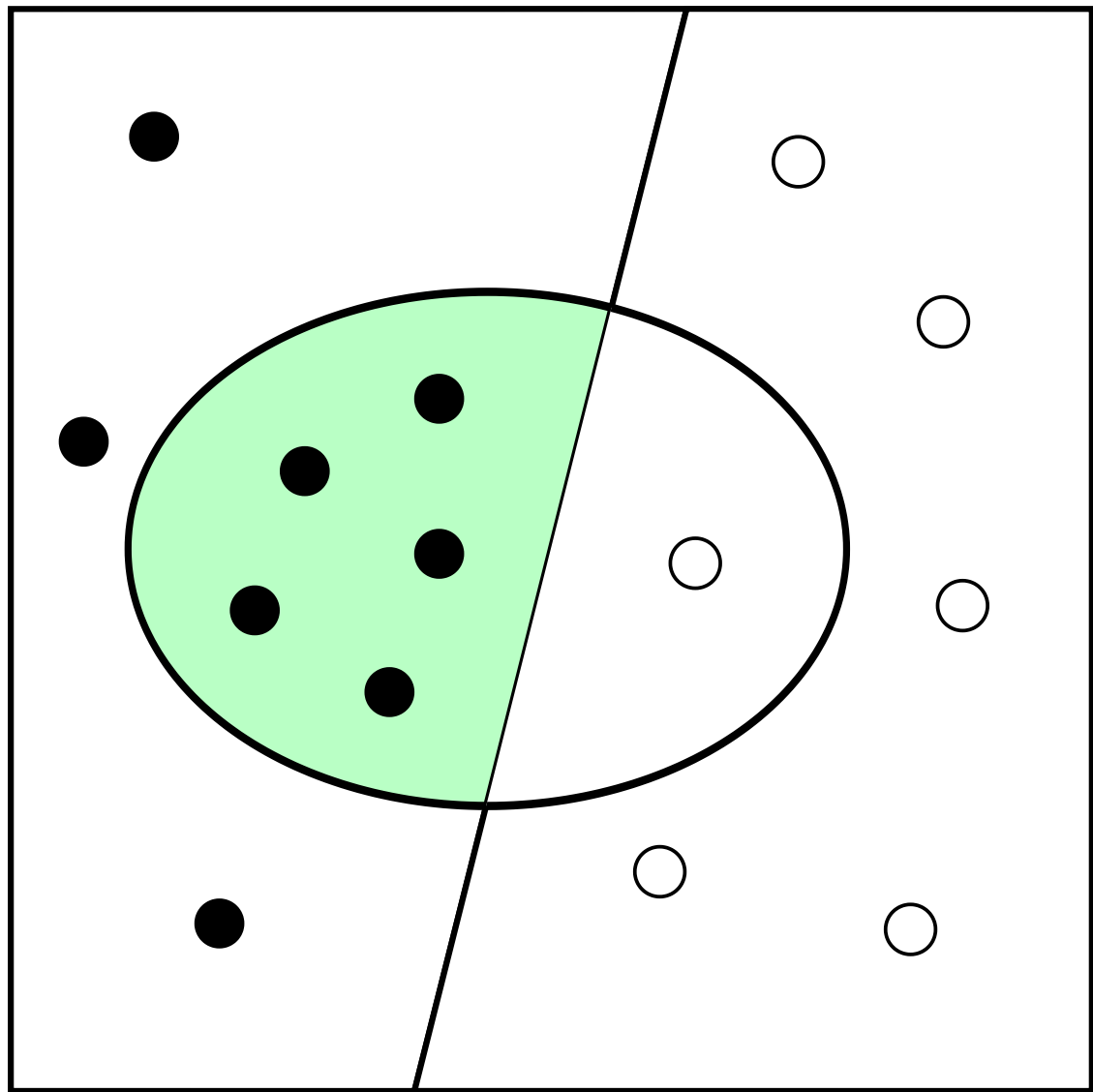


- breast cancer
- no breast cancer
- prediction breast cancer

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

"yes" = cancer present
"no" = cancer not present



● breast cancer

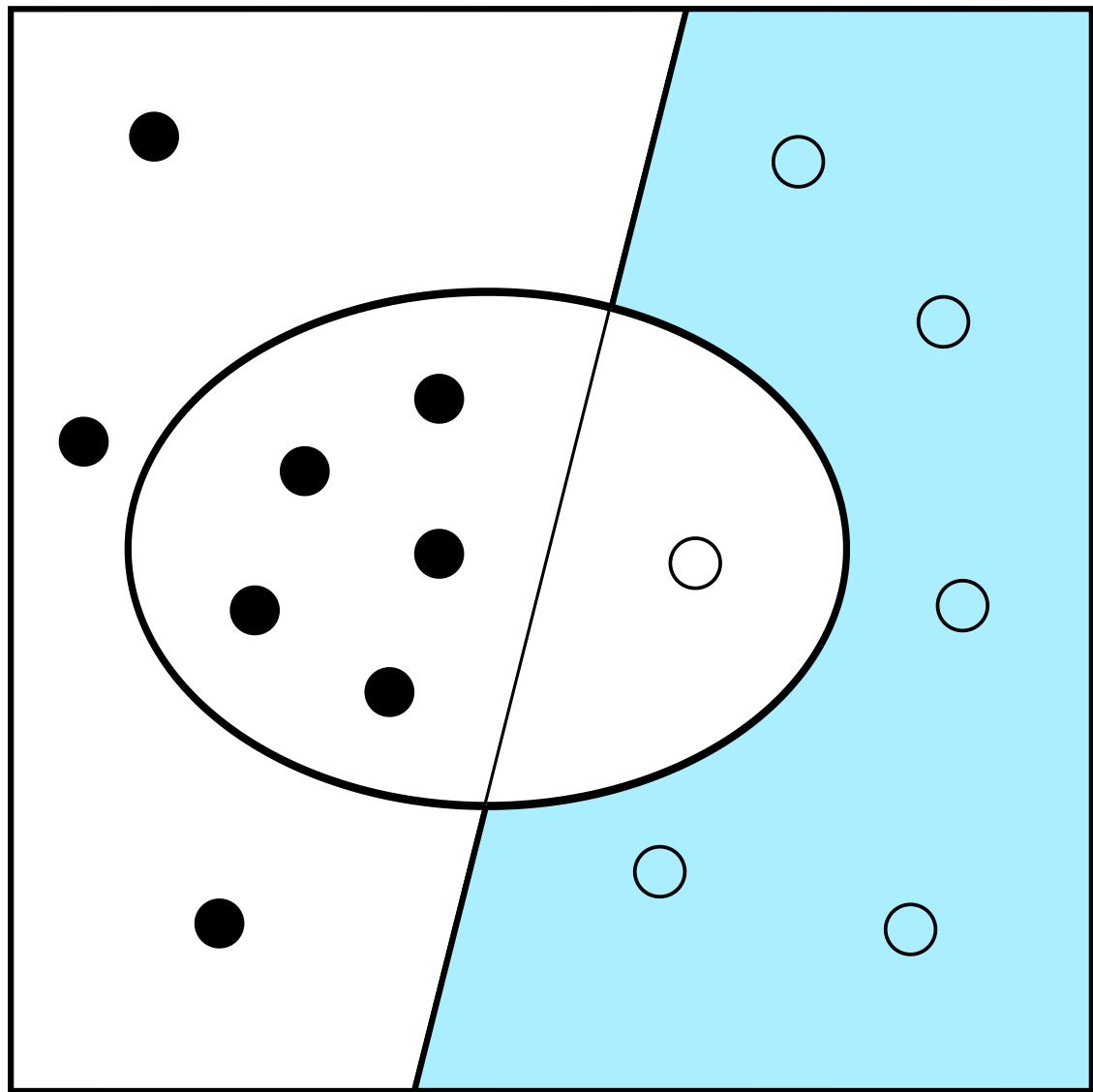
○ no breast cancer

○ prediction breast cancer

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

"yes" = cancer present
"no" = cancer not present



● breast cancer

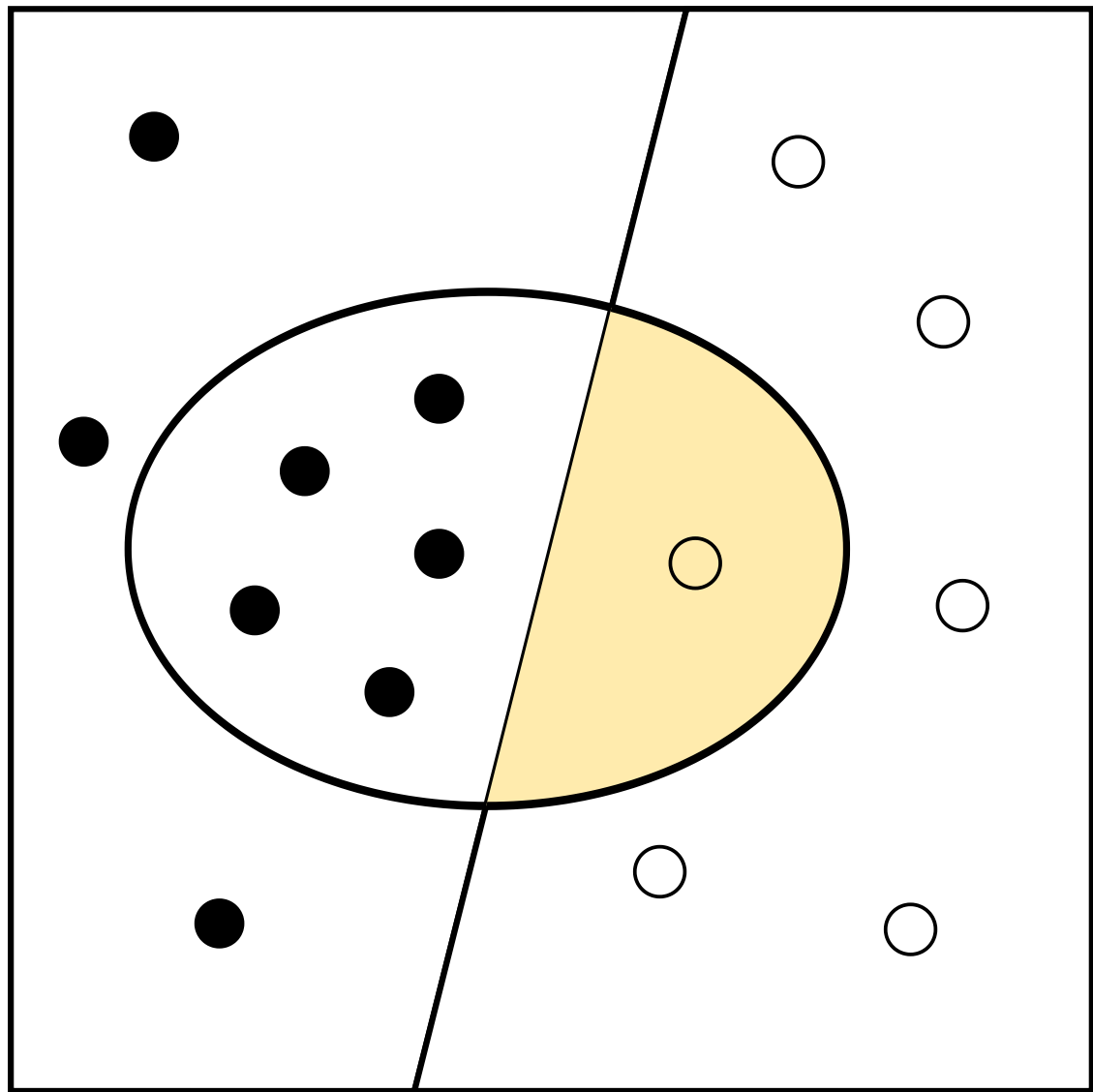
○ no breast cancer

○ prediction
breast cancer

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

"yes" = cancer present
"no" = cancer not present

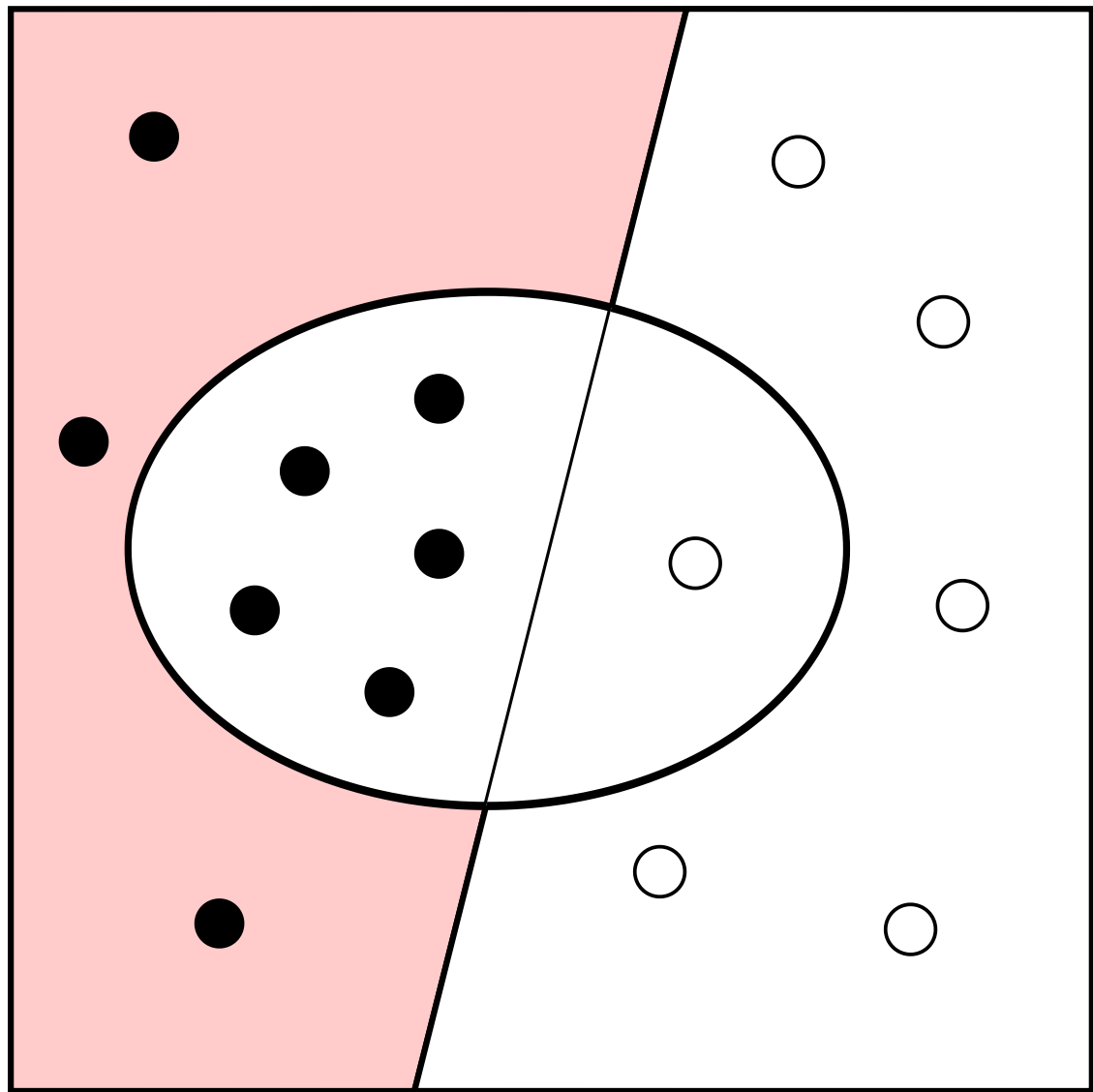


- breast cancer
- no breast cancer
- prediction breast cancer

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

"yes" = cancer present
"no" = cancer not present

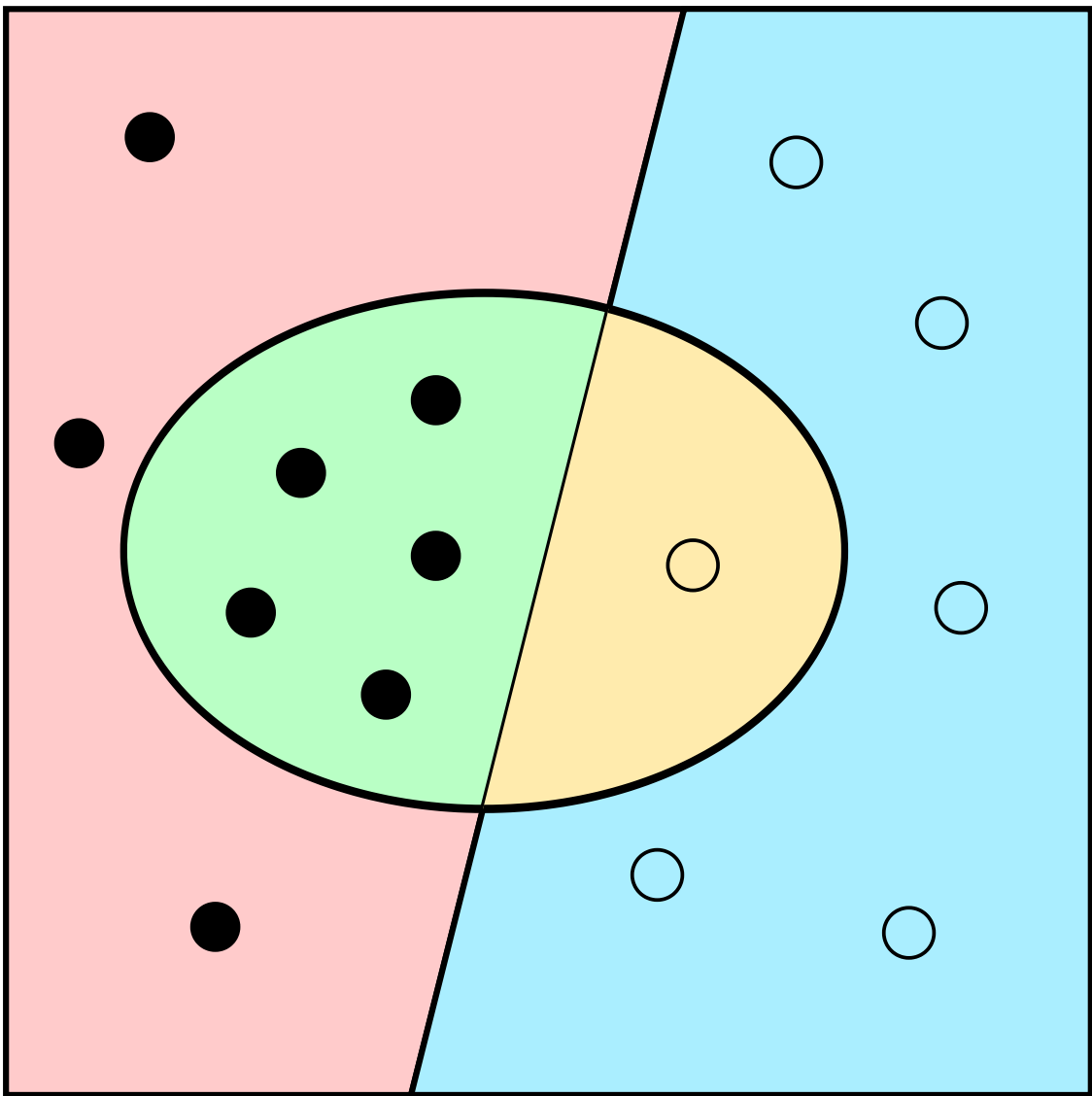


- breast cancer
- no breast cancer
- prediction breast cancer

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

2x2 table summarizing the true and predicted values of observations.



- breast cancer
- no breast cancer
- prediction breast cancer

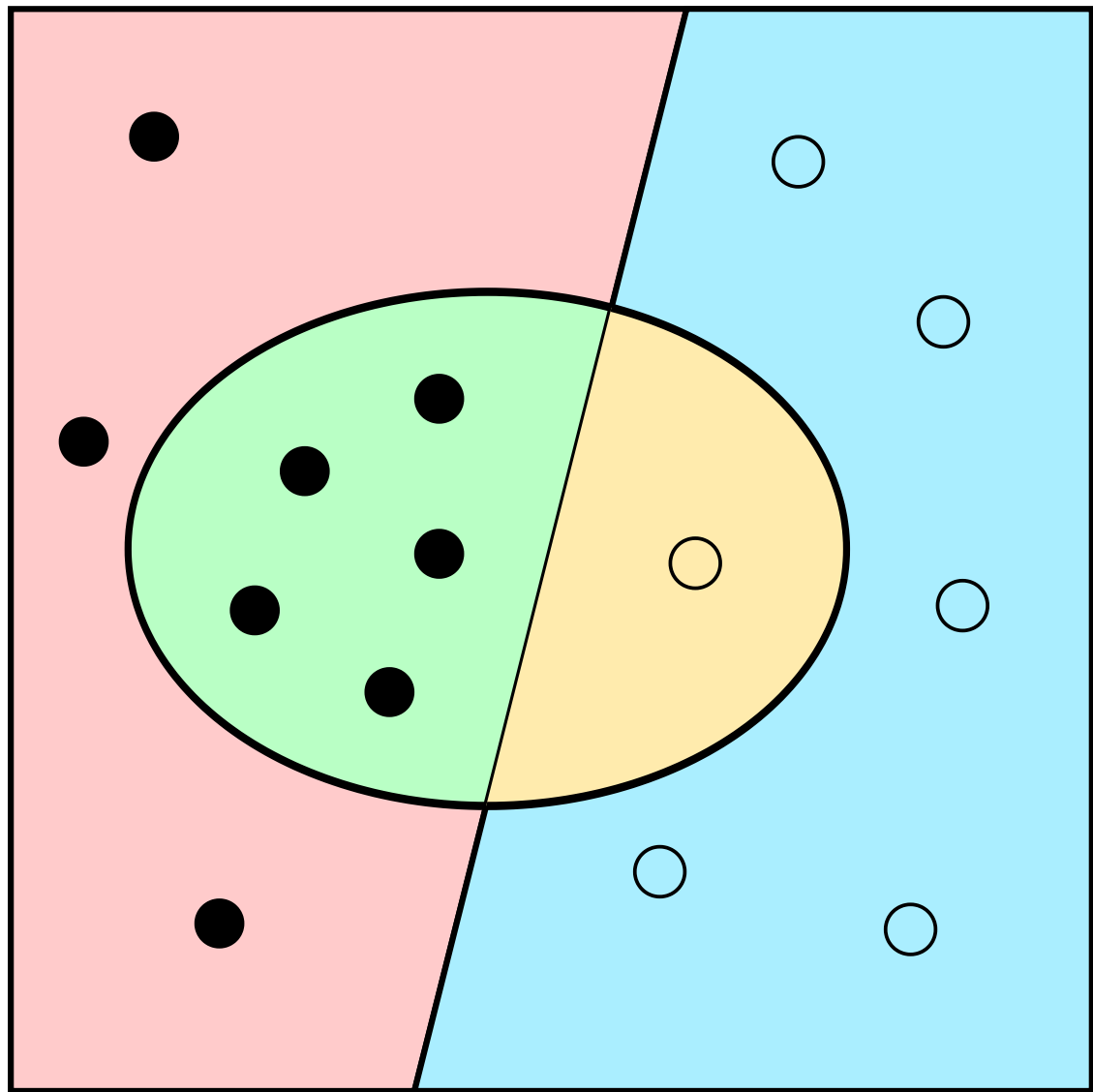
		PREDICTED VALUE		TOTAL
		YES	NO	
TRUE VALUE	YES	true positive	false negative	
	NO	false positive	true negative	
TOTAL				

"yes" = cancer present
"no" = cancer not present

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

2x2 table summarizing the true and predicted values of observations.



● breast cancer

○ no breast cancer

○ prediction breast cancer

		PREDICTED VALUE		TOTAL
		YES	NO	
TRUE VALUE	YES	5	3	8
	NO	1	5	6
TOTAL		6	8	14

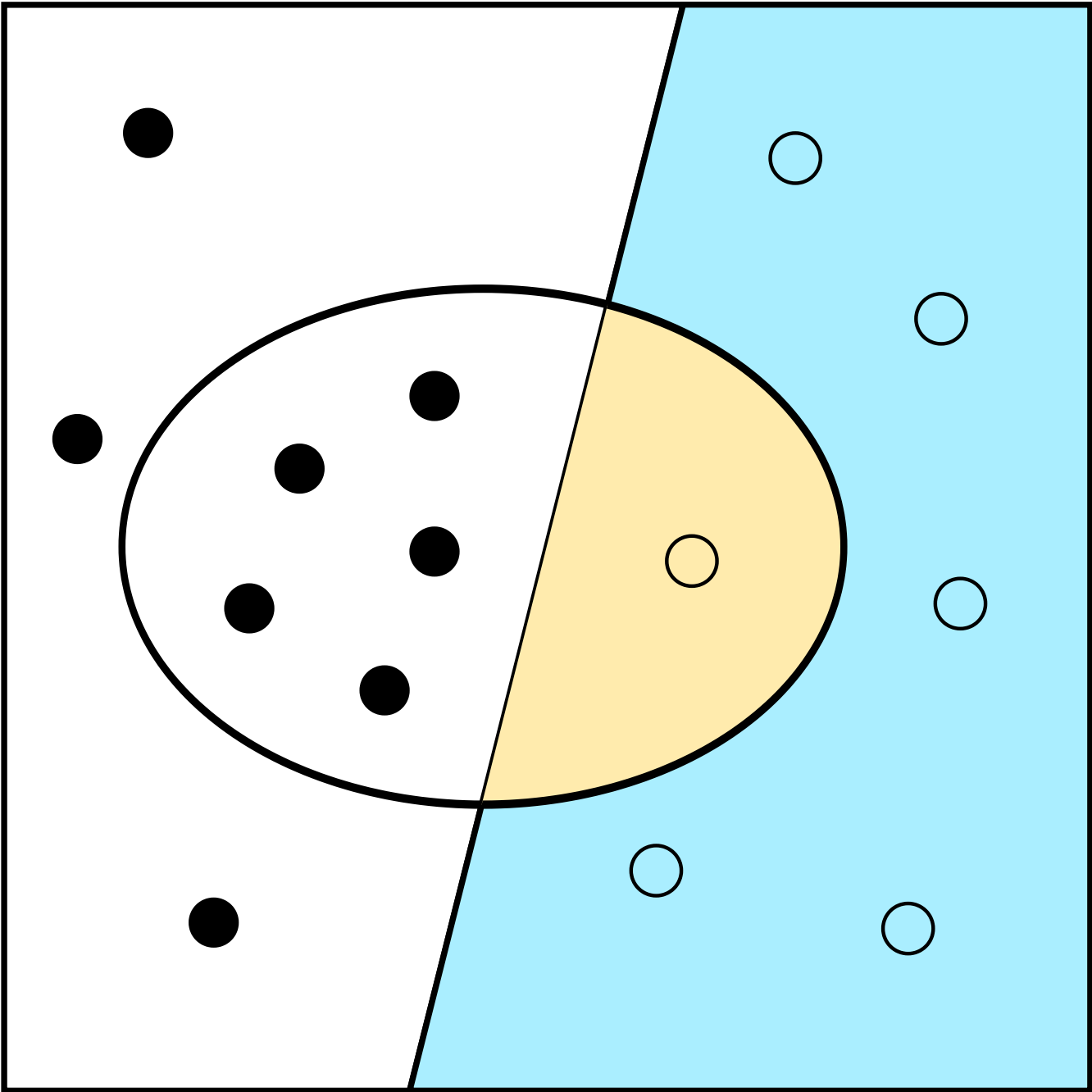
"yes" = cancer present
"no" = cancer not present

TRUE VALUE	PREDICTED VALUE
yes	yes
no	no
no	yes
yes	yes
no	no
yes	no
no	no
yes	yes
yes	no
yes	no
yes	yes
no	no
yes	yes
no	no

Confusion Matrix

2x2 table summarizing the true and predicted values of observations.

Calculate the
false positive rate: $1 / (1 + 5) = 16.6\%$

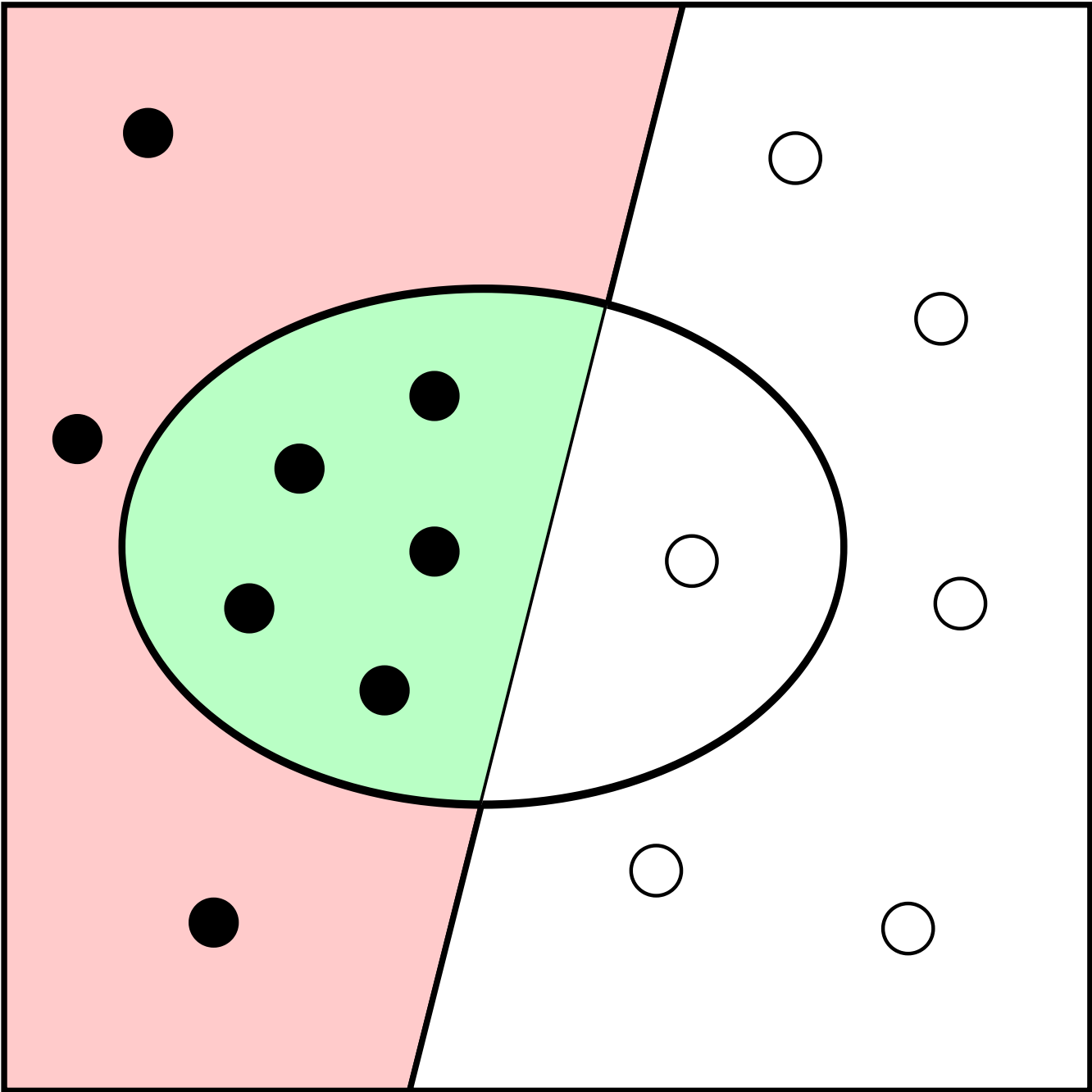


		PREDICTED VALUE		
		YES	NO	TOTAL
TRUE VALUE	YES	5	3	8
	NO	1	5	6
TOTAL		6	8	14

Confusion Matrix

2x2 table summarizing the true and predicted values of observations.

Calculate the
false positive rate: $1 / (1 + 5) = 16.6\%$
false negative rate: $3 / (3 + 5) = 37.5\%$



		PREDICTED VALUE		TOTAL
		YES	NO	
TRUE VALUE	YES	5	3	8
	NO	1	5	6
TOTAL		6	8	14

Confusion Matrix

2x2 table summarizing the true and predicted values of observations.

Calculate the

false positive rate: $1 / (1 + 5) = 16.6\%$

false negative rate: $3 / (3 + 5) = 37.5\%$

There's a significant imbalance between false positive and false negative errors!

		PREDICTED VALUE		TOTAL
		YES	NO	
TRUE VALUE	YES	5	3	8
	NO	1	5	6
TOTAL		6	8	14

Confusion Matrix

2x2 table summarizing the true and predicted values of observations.

Calculate the
false positive rate: $1 / (1 + 5) = 16.6\%$
false negative rate: $3 / (3 + 5) = 37.5\%$

There's a significant imbalance between false positive and false negative errors! Which error is more important?

		PREDICTED VALUE		TOTAL
		YES	NO	
TRUE VALUE	YES	5	3	8
	NO	1	5	6
TOTAL		6	8	14

ethical aspects
of AI

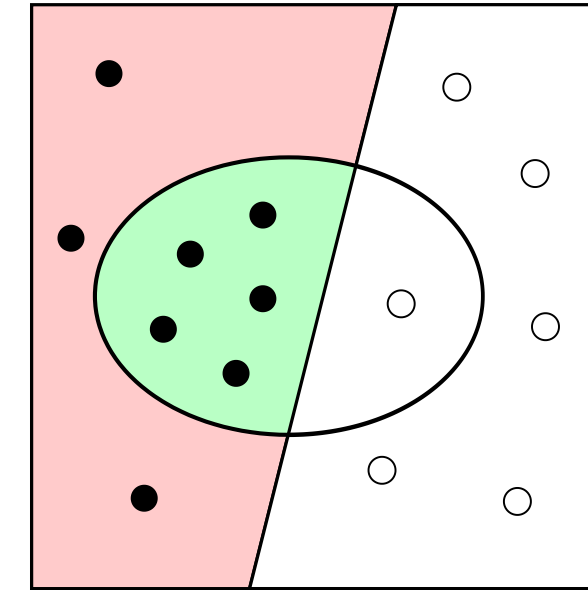
economic
aspects of AI

legal aspects
of AI

Different Performance Indicators

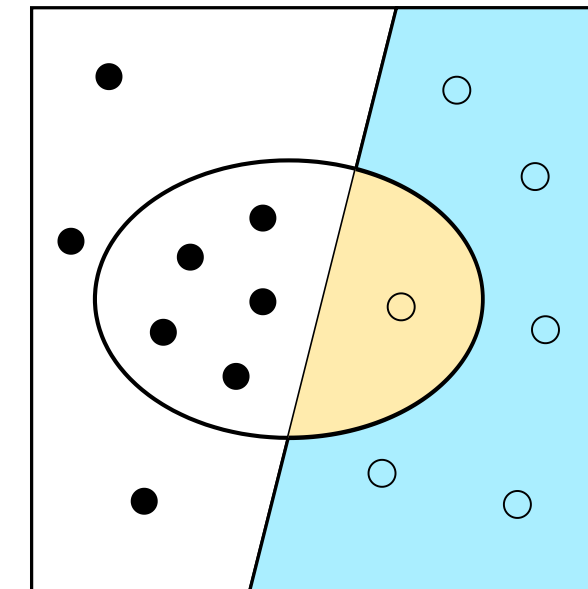
recall
(sensitivity) $\frac{t_p}{t_p + f_n}$

The probability with which a positive observation is correctly classified as positive.



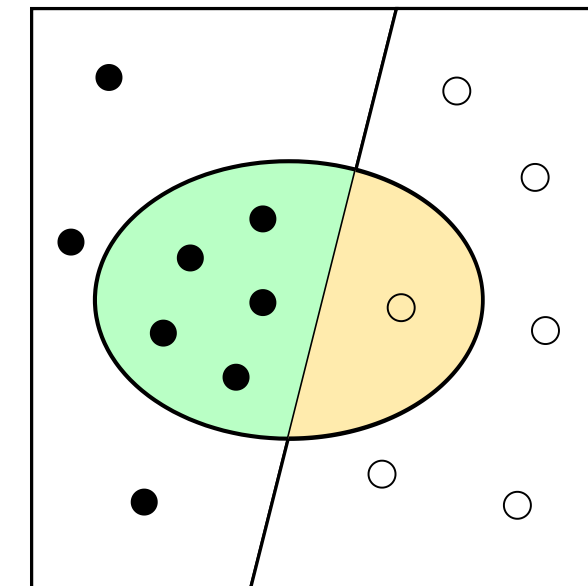
specificity
 $\frac{t_n}{t_n + f_p}$

The probability with which a negative observation is correctly classified as negative.



precision
(pos. pred. value) $\frac{t_p}{t_p + f_p}$

The percentage of observations correctly classified as positive in the total of observations classified as positive.



		PRED.	
		YES	NO
TRUE	YES	t_p	f_n
	NO	f_p	t_n

Combined Measures

The F -score (also F_1 -score) is the harmonic mean of precision and recall (sensitivity), where both values are weighted equally.

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

General: The F_β -score with different weights β that scale the importance of recall.

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}$$

For F_2 recall is twice as important as precision.

For $F_{0.5}$ recall is half as important as precision.

Class Frequency

Example: *Breast cancer screening program. One in 100 women actually has breast cancer. This means: negative cases are overrepresented 1:100.*

- A classifier that always predicts "no breast cancer" has an accuracy of 99%!
- However, it has a false positive rate of 0% and a false negative rate of 100%.
- If only accuracy is reported, the classifier looks very good. In reality, however, it's absolutely useless for the application.

When to Use Which Performance Indicator?

- **Accuracy:** When **classes are equally frequent** and false positive and false negative errors are equally severe.
- **Recall (sensitivity):** When classes are not equally frequent and **false negative errors** are particularly severe.
- **Precision** (positive predictive value) or **specificity:** When classes are not equally frequent and **false positive errors** are particularly severe.
- **F_1 :** When **classes are not equally frequent** and both false positive and false negative errors are severe.

Summary

- Knowing a classifier's **performance** is crucial for **guiding training** and assessing the classifier's **effectiveness**.
- Classifiers make **different types of errors**:
 - **False positive**: a negative observation is classified as positive.
 - **False negative**: a positive observation is classified as negative.
- Depending on the importance of different errors and the class balance, **various performance indicators** are used:
 - Accuracy
 - Precision (positive predictive value), recall (sensitivity), specificity
 - F_1 or F_β Score

Exercise (5 min)

For the true and predicted values in the table on the right:

- Count the false positive, false negative, true positive, and true negative predictions.
- Calculate the error rate and accuracy (= 1 - error rate), precision, recall, and specificity.
- Calculate the F_1 Score.
- For which tasks is this classifier well suited or not well suited?

TRUE VALUE	PREDICTED VALUE
yes	yes
no	yes
yes	yes
yes	yes
yes	yes
no	no
no	no
yes	yes
no	yes
no	no
yes	no
no	yes
yes	yes
no	yes

Solution

- $f_p = 4; \quad f_n = 1; \quad t_p = 6; \quad t_n = 3$
- Error rate: $5/14 = 0.36$
Accuracy: $1 - 0.36 = 0.64$
Recall: $r_p / (t_p + f_n) = 6 / (6 + 1) = 0.86$
Specificity: $t_n / (t_n + f_p) = 3 / (3 + 4) = 0.42$
Precision: $t_p / (t_p + f_p) = 6 / (6 + 4) = 0.6$
- F_1 -Score: $2 * (precision * recall) / (precision + recall)$
 $= 2 * (0.6 * 0.86) / (0.6 + 0.86) = 0.71$
- This classifier is well-suited for tasks where false positives are not critical. For example pre-selection of social media posts that violate community guidelines that are then reviewed by humans.

TRUE VALUE	PREDICTED VALUE
yes	yes
no	yes
yes	yes
yes	yes
yes	yes
no	no
no	no
yes	yes
no	yes
no	no
yes	no
no	yes
yes	yes
no	yes